

Variation in Many-Analyst Projects: Harmful or Helpful? A Multiverse Approach to Address Different Sources of Variation

Thesis Proposal

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1 Introduction

Recent studies have shown considerable variation in outcomes when multiple researchers answer the same question using the same data, raising concerns about the reproducibility of science (e.g., Hoozeven et al., 2023; Silberzahn et al., 2018). Kowall et al. (2025) asked 16 teams “Does marital status (cohabiting married versus never married) influence the incidence of cardiovascular disease?”. Their decisions varied widely (Table 1), resulting in a range of negative to positive effects.

Table 1: Different choices that were made by analysts in the study by Kowall et al. (2025)

(a) Analytical decisions about data processing and statistical models.

Missing Data Handling Methods	Type of Model
Complete cases (16)	Cox Proportional Hazards model (6)
Imputed data (3)	Logistic Regression (5)
“No information” (1)	Log-binomial model (2)
	Mixed-effects logistic regression (1)
	Discrete time mixed effect model (1)
	Generalized estimating equations (1)

(b) Substantive decisions about variable definitions and target populations.

Outcome (Dependent Variable)	Adjustment Set (Covariates)	Type of Data (Change Over Time)	Exclusion Criteria
Heart attack / stroke ever (16)	Age (12)	Strictly longitudinal (10)	Subjects married before age 18
Age at heart attack / stroke (9)	Sex (12)	Longitudinal with wave 1 as the only baseline (3)	Subjects from Israel
Heart attack / stroke since last interview (7)	Country (7)	Strictly or partly cross-sectional (6)	Persons under age 55 at recruitment
Main cause of death heart attack / stroke (5)	Smoking (6)		Subjects born after 1954
	Education (4)		
	Physical activity (4)		Change in exposure

Note: The numbers in brackets behind the decisions denote how many analysts made this choice in their analysis. The outcome and adjustment sets were composed of combinations of the variables in this table.

Table 2: Sources of variation according to Auspurg and Brüderl (2024)

Harmful	Helpful
(a) Model uncertainty, including subjective choices in (missing) data handling, and choosing a type of statistical model	(c) Substantively meaningful variation, based on different definitions of dependent, independent and co-variables and/or target populations (e.g., different exclusion criteria or changes over time)
(b) Coding Errors	

Many-analyst projects can reveal natural variation in science, but proper explanations are lacking. Prior beliefs, level of experience and analysis quality do not seem explanatory (Silberzahn et al., 2018). Some attribute variation to different variable definitions (Hoogeveen et al., 2023), others link it to data type (Kowall et al., 2025). These choices reflect different causal assumptions, indicating some decisions may be theoretical rather than analytical (Bulbulia, 2023; Del Giudice & Gangestad, 2021; Rohrer et al., 2025). Therefore, interpreting variation within and across studies remains challenging – is variation harmful to reproducibility or helpful for theory development? Auspurg and Brüderl (2024) argue that distinguishing sources of variation is relevant to answer this question (Table 2), leading to my research question.

Research Question To what extent do different types of analysis decisions/sources of variation, including handling missing data, choice of statistical model and definitions of outcome variables, covariates and target population, contribute to variation in the outcomes of many-analyst projects?

Expectations I expect substantial variation across analytical choices, like in previous studies. Given their diverse explanations, I have reservations about making predictions about what sources will contribute the most. As I will base my study on Kowall et al. (2025), similar patterns may arise, although I assume this with great caution, as my analysis will be more extensive, possibly revealing different patterns.

2 Analytic Strategy

Original Many-Analyst Study Because Kowall et al. (2025) included multiple sources of variation, I will focus on analysts’ choices in this study (Table 1). A few notes:

- As exposure, all teams but two used “cohabiting married” (CM). The exceptions also included registered partnership, and married living apart, answering different questions, I will use only CM.
- As outcome, all teams used different combinations of the four variables in Table 1.
- The adjustment sets in the original study contained 25 variables. To ensure my analysis is feasible, I will only include covariates used by at least four analysts, except the ‘no covariates’ category which will also be included. I am aware this is an arbitrary decision.
- Analysts used different exclusion criteria and data types, implying they also had different sample sizes, which Kowall et al. (2025) did not explicitly name as source of variation, but may obviously have resulted in different estimates.

Data Like Kowall et al. (2025), I will use the Survey of Health, Ageing and Retirement in Europe (SHARE): a freely available (after registration), panel study conducted in 28 European countries in different waves (in the original study, wave 1-7 were included).

Multiverse Analysis Many-analyst projects show how analytic decisions affect results, but not systematically. Conversely, multiverse analysis does this by systematically mapping all reasonable research paths that result from different combinations of analytic choices (Steege et al., 2016). This method allows distinguishing different sources of variation and evaluating their impact. Figure 1 shows an overview of the steps I will follow in my multiverse (Simson et al., 2024).

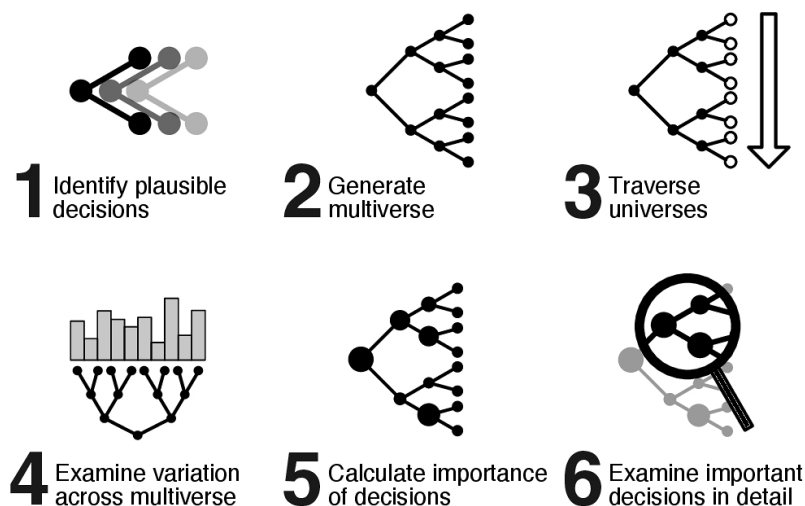


Figure 1: Comprehensive Overview of Multiverse Analysis, from Simson et al. (2024)

Table 3: Number of possible analysis paths based on various decision patterns.

Decision	Unique possibilities	Number of combinations
Missingness	2	2
Model Type	6	6
Outcome	4	$2^4 = 16$
Adjustment Set	7	$2^7 = 128$
Data Type	3	3
Exclusion	4	4
Total		294.912

Note: As explained above, the outcome variable was made up by making combinations of one or more out of four variables related to cardiovascular outcomes.

Note: The exclusion criterium ‘change of exposure’ consisted of multiple levels, which is why I decided not to include it for now.

Decision Space First, I will determine my *decision space*, including all reasonable research paths, by systematically combining all decisions made by analysts in Kowall et al. (2025). However, Figure 2 and Table 3 show that — even when looking at a small part of the multiverse — the number of different paths can quickly become very large.

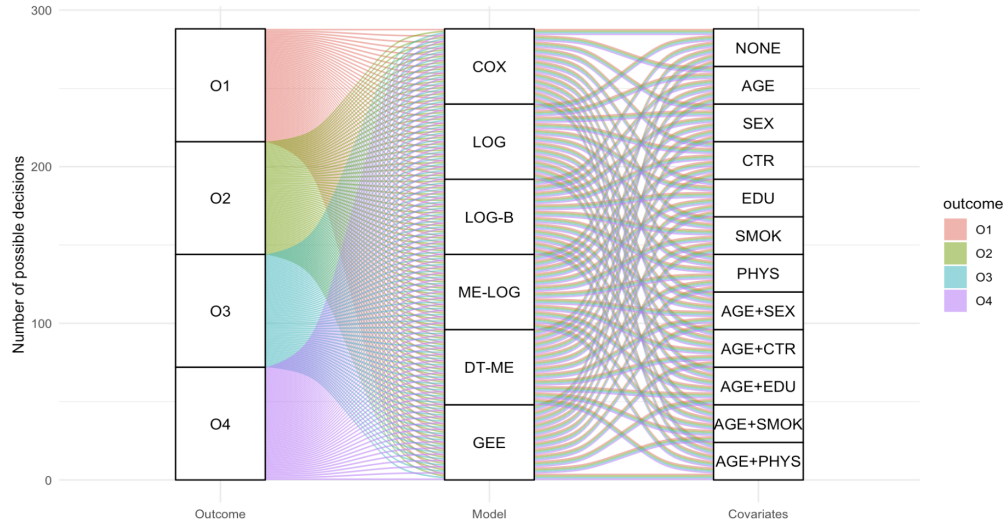


Figure 2: Simplified overview of the multiverse analysis in this study. This figure shows only a part of the multiverse; many more combinations of the outcome variables and adjustment sets are possible and other decisions will also be included. *O1-O4: outcome variables, as described in Table 1; COX: Cox Proportional Hazards model, LOG: logistic regression, LOG-B: log-binomial model, ME-LOG: mixed-effects logistic regression, DT-ME: discrete time mixed effect model, GEE: generalized estimating equations; CTR: country, EDU: education, SMOK: smoking, PHYS: physical activity.*

Sampling and Approximation I will base my analysis on an approximation algorithm by Liu et al. (2023), where research paths are sampled from the decision space, and mean effects and decision sensitivity are calculated intermediately — with sample sizes based on computational and time limits. They quantify sensitivity using a K-samples Anderson–Darling test — a non-parametric method that simultaneously evaluates whether the outcomes of k different research paths follow the same distributions, assuming that sensitive decisions lead to wider variety in distributions. Additionally, they use bootstrapping to estimate 95%-confidence intervals around the sensitivity measures.

Furthermore, the distribution of the estimates of the sampled multiverse will be compared to that of the estimates in the original study to see whether the multiverse captures the same variation as the many-analyst project.

Software Analyses will be conducted in RStudio or Jupyter (Liu et al. (2023) provided python-code).

Table 4: If the analyses are performed in RStudio (version 4.4.2/4.4.3) (Team, n.d.), I wil likely use (at least) the following packages:

Package	Citation
tidyverse	(Wickham et al., 2019)
ggalluvial	(Brunson, 2020)
multiverse	(Sarma et al., 2021)
specr	(Masur & Scharrow, 2020)
lme4	(Bates et al., 2015)
survival	(Therneau, 2024)
geepack	(Halekoh et al., 2006)

References Used in Text

- Auspurg, K., & Brüderl, J. (2024). Toward a more credible assessment of the credibility of science by many-analyst studies. *Proceedings of the National Academy of Sciences*, 121(38), e2404035121. <https://doi.org/10.1073/pnas.2404035121>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using **lme4**. *Journal of Statistical Software*, 67(1). <https://doi.org/10.18637/jss.v067.i01>
- Brunson, J. (2020). Ggalluvial: Layered Grammar for Alluvial Plots. *Journal of Open Source Software*, 5(49), 2017. <https://doi.org/10.21105/joss.02017>
- Bulbulia, J. A. (2023). A workflow for causal inference in cross-cultural psychology. *Religion, Brain & Behavior*, 13(3), 291–306. <https://doi.org/10.1080/2153599X.2022.2070245>
- Del Giudice, M., & Gangestad, S. W. (2021). A Traveler’s Guide to the Multiverse: Promises, Pitfalls, and a Framework for the Evaluation of Analytic Decisions. *Advances in Methods and Practices in Psychological Science*, 4(1), 2515245920954925. <https://doi.org/10.1177/2515245920954925>
- Halekoh, U., Højsgaard, S., & Yan, J. (2006). The R Package **geepack** for Generalized Estimating Equations. *Journal of Statistical Software*, 15(2). <https://doi.org/10.18637/jss.v015.i02>
- Hoogeveen, S., Sarafoglou, A., Van Elk, M., & Wagenmakers, E.-J. (2023). Many-analysts religion project: Reflection and conclusion. *Religion, Brain & Behavior*, 13(3), 356–363. <https://doi.org/10.1080/2153599X.2022.2070263>
- Kowall, B., Ahrenfeldt, L. J., Basten, J., Becher, H., Brand, T., Braun, J., Casjens, S., Claessen, H., Denz, R., Diebner, H. H., Diexer, S., Eisemann, N., Furrer, E., Galetzka, W., Girschik, C., Karch, A., Mikolajczyk, R., Peters, M., Rospleszcz, S., . . . Rübsamen, N. (2025). Marital status and risk of cardiovascular disease – a multi-analyst study in epidemiology. *European Journal of Epidemiology*, 40(5), 497–509. <https://doi.org/10.1007/s10654-025-01235-8>
- Liu, Y., Althoff, T., & Heer, J. (2023, May). Approximation and Progressive Display of Multiverse Analyses [arXiv:2305.08323 [cs]]. <https://doi.org/10.48550/arXiv.2305.08323>
- Masur, P. K., & Scharkow, M. (2020, March). Specr: Conducting and Visualizing Specification Curve Analyses [Institution: Comprehensive R Archive Network Pages: 1.0.0]. <https://doi.org/10.32614/CRAN.package.specr>
- Rohrer, J. M., Smith, G. D., & Munafò, M. (2025). What can be learned when multiple analysts arrive at different estimates. *European Journal of Epidemiology*, 40(5), 493–495. <https://doi.org/10.1007/s10654-025-01249-2>
- Sarma, A., Kale, A., Moon, M. J., Taback, N., Chevalier, F., Hullman, J., & Kay, M. (2021, April). Multiverse: Multiplexing Alternative Data Analyses in R Notebooks. <https://doi.org/10.31219/osf.io/yfbwm>
- Silberzahn, R., Uhlmann, E. L., Martin, D. P., Anselmi, P., Aust, F., Awtrey, E., Bahník, Š., Bai, F., Bannard, C., Bonnier, E., Carlsson, R., Cheung, F., Christensen, G., Clay, R., Craig, M. A., Dalla Rosa, A., Dam, L., Evans, M. H., Flores Cervantes, I., . . . Nosek, B. A. (2018). Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results. *Advances in Methods and Practices in Psychological Science*, 1(3), 337–356. <https://doi.org/10.1177/2515245917747646>
- Simson, J., Pfisterer, F., & Kern, C. (2024). One Model Many Scores: Using Multiverse Analysis to Prevent Fairness Hacking and Evaluate the Influence of Model Design Decisions. *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 1305–1320. <https://doi.org/10.1145/3630106.3658974>

- Steege, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*, 11(5), 702–712. <https://doi.org/10.1177/1745691616658637>
- Team, R. C. (n.d.). R: A Language and Environment for Statistical Computing. <https://www.R-project.org/>
- Therneau, T. (2024). *A Package for Survival Analysis in R*. R package. <https://CRAN.R-project.org/package=survival>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., ... Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>

Planned References for Research Report (including references used in this proposal)

- Aczel, B., Szaszi, B., Nilsson, G., Van Den Akker, O. R., Albers, C. J., Van Assen, M. A., Bastiaansen, J. A., Benjamin, D., Boehm, U., Botvinik-Nezer, R., Bringmann, L. F., Busch, N. A., Caruyer, E., Cataldo, A. M., Cowan, N., Delios, A., Van Dongen, N. N., Donkin, C., Van Doorn, J. B., ... Wagenmakers, E.-J. (2021). Consensus-based guidance for conducting and reporting multi-analyst studies. *eLife*, 10, e72185. <https://doi.org/10.7554/eLife.72185>
- Auspurg, K., & Brüderl, J. (2024). Toward a more credible assessment of the credibility of science by many-analyst studies. *Proceedings of the National Academy of Sciences*, 121(38), e2404035121. <https://doi.org/10.1073/pnas.2404035121>
- Bastiaansen, J. A., Kunkels, Y. K., Blaauw, F. J., Boker, S. M., Ceulemans, E., Chen, M., Chow, S.-M., De Jonge, P., Emerencia, A. C., Epskamp, S., Fisher, A. J., Hamaker, E. L., Kuppens, P., Lutz, W., Meyer, M. J., Moulder, R., Oravecz, Z., Riese, H., Rubel, J., ... Bringmann, L. F. (2020). Time to get personal? The impact of researchers choices on the selection of treatment targets using the experience sampling methodology. *Journal of Psychosomatic Research*, 137, 110211. <https://doi.org/10.1016/j.jpsychores.2020.110211>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using **lme4**. *Journal of Statistical Software*, 67(1). <https://doi.org/10.18637/jss.v067.i01>
- Bloom, P. A., VanTieghem, M., Gabard-Durnam, L., Gee, D. G., Flannery, J., Caldera, C., Goff, B., Telzer, E. H., Humphreys, K. L., Fareri, D. S., Shapiro, M., Algharazi, S., Bolger, N., Aly, M., & Tottenham, N. (2022). Age-related change in task-evoked amygdala—prefrontal circuitry: A multiverse approach with an accelerated longitudinal cohort aged 4–22 years. *Human Brain Mapping*, 43(10), 3221–3244. <https://doi.org/10.1002/hbm.25847>
- Breznau, N., Rinke, E. M., Wuttke, A., Nguyen, H. H. V., Adem, M., Adriaans, J., Alvarez-Benjumea, A., Andersen, H. K., Auer, D., Azevedo, F., Bahnsen, O., Balzer, D., Bauer, G., Bauer, P. C., Baumann, M., Baute, S., Benoit, V., Bernauer, J., Berning, C., ... Zóltak, T. (2022). Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. *Proceedings of the National Academy of Sciences*, 119(44), e2203150119. <https://doi.org/10.1073/pnas.2203150119>
- Brunson, J. (2020). Ggalluvial: Layered Grammar for Alluvial Plots. *Journal of Open Source Software*, 5(49), 2017. <https://doi.org/10.21105/joss.02017>
- Bulbulia, J. A. (2023). A workflow for causal inference in cross-cultural psychology. *Religion, Brain & Behavior*, 13(3), 291–306. <https://doi.org/10.1080/2153599X.2022.2070245>

- Del Giudice, M., & Gangestad, S. W. (2021). A Traveler’s Guide to the Multiverse: Promises, Pitfalls, and a Framework for the Evaluation of Analytic Decisions. *Advances in Methods and Practices in Psychological Science*, 4(1), 2515245920954925. <https://doi.org/10.1177/2515245920954925>
- Girardi, P., Vesely, A., Lakens, D., Altoè, G., Pastore, M., Calcagni, A., & Finos, L. (2023, October). Post-selection Inference in Multiverse Analysis (PIMA): An inferential framework based on the sign flipping score test [arXiv:2210.02794 [stat]]. <https://doi.org/10.48550/arXiv.2210.02794>
Comment: 37 pages, 2 figures.
- Götz, M., Sarma, A., & O’Boyle, E. H. (2024). The multiverse of universes: A tutorial to plan, execute and interpret multiverses analyses using the R package *multiverse*. *International Journal of Psychology*, 59(6), 1003–1014. <https://doi.org/10.1002/ijop.13229>
- Gould, E., Fraser, H. S., Parker, T. H., Nakagawa, S., Griffith, S. C., Vesk, P. A., Fidler, F., Hamilton, D. G., Abbey-Lee, R. N., Abbott, J. K., Aguirre, L. A., Alcaraz, C., Aloni, I., Altschul, D., Arekar, K., Atkins, J. W., Atkinson, J., Baker, C. M., Barrett, M., . . . Zitomer, R. A. (2025). Same data, different analysts: Variation in effect sizes due to analytical decisions in ecology and evolutionary biology. *BMC Biology*, 23(1), 35. <https://doi.org/10.1186/s12915-024-02101-x>
- Halekoh, U., Højsgaard, S., & Yan, J. (2006). The R Package **geepack** for Generalized Estimating Equations. *Journal of Statistical Software*, 15(2). <https://doi.org/10.18637/jss.v015.i02>
- Hanel, P. H. P., & Zarzeczna, N. (2023). From multiverse analysis to multiverse operationalisations: 262,143 ways of measuring well-being. *Religion, Brain & Behavior*, 13(3), 309–313. <https://doi.org/10.1080/2153599X.2022.2070259>
- Hoogeveen, S., Sarafoglou, A., Aczel, B., Aditya, Y., Alayan, A. J., Allen, P. J., Altay, S., Alzahawi, S., Amir, Y., Anthony, F.-V., Kwame Appiah, O., Atkinson, Q. D., Baimel, A., Balkaya-Ince, M., Balsamo, M., Banker, S., Bartoš, F., Becerra, M., Beffara, B., . . . Wagenmakers, E.-J. (2023). A many-analysts approach to the relation between religiosity and well-being. *Religion, Brain & Behavior*, 13(3), 237–283. <https://doi.org/10.1080/2153599X.2022.2070255>
- Hoogeveen, S., Sarafoglou, A., Van Elk, M., & Wagenmakers, E.-J. (2023). Many-analysts religion project: Reflection and conclusion. *Religion, Brain & Behavior*, 13(3), 356–363. <https://doi.org/10.1080/2153599X.2022.2070263>
- Kowall, B., Ahrenfeldt, L. J., Basten, J., Becher, H., Brand, T., Braun, J., Casjens, S., Claessen, H., Denz, R., Diebner, H. H., Diexer, S., Eisemann, N., Furrer, E., Galetzka, W., Girschik, C., Karch, A., Mikolajczyk, R., Peters, M., Rospleszcz, S., . . . Rübsamen, N. (2025). Marital status and risk of cardiovascular disease – a multi-analyst study in epidemiology. *European Journal of Epidemiology*, 40(5), 497–509. <https://doi.org/10.1007/s10654-025-01235-8>
- Kummerfeld, E., & Jones, G. L. (2023). One data set, many analysts: Implications for practicing scientists. *Frontiers in Psychology*, 14, 1094150. <https://doi.org/10.3389/fpsyg.2023.1094150>
- Liu, Y., Althoff, T., & Heer, J. (2023, May). Approximation and Progressive Display of Multiverse Analyses [arXiv:2305.08323 [cs]]. <https://doi.org/10.48550/arXiv.2305.08323>
- Masur, P. K., & Scharnow, M. (2020, March). Specr: Conducting and Visualizing Specification Curve Analyses [Institution: Comprehensive R Archive Network Pages: 1.0.0]. <https://doi.org/10.32614/CRAN.package.specr>

- Parsons, S. (2022). Exploring reliability heterogeneity with multiverse analyses: Data processing decisions unpredictably influence measurement reliability. *Meta-Psychology*, 6. <https://doi.org/10.15626/MP.2020.2577>
- Patel, C. J., Burford, B., & Ioannidis, J. P. (2015). Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. *Journal of Clinical Epidemiology*, 68(9), 1046–1058. <https://doi.org/10.1016/j.jclinepi.2015.05.029>
- Rohrer, J. M., Smith, G. D., & Munafò, M. (2025). What can be learned when multiple analysts arrive at different estimates. *European Journal of Epidemiology*, 40(5), 493–495. <https://doi.org/10.1007/s10654-025-01249-2>
- Salganik, M. J., Lundberg, I., Kindel, A. T., Ahearn, C. E., Al-Ghoneim, K., Almaatouq, A., Altschul, D. M., Brand, J. E., Carnegie, N. B., Compton, R. J., Datta, D., Davidson, T., Filippova, A., Gilroy, C., Goode, B. J., Jahani, E., Kashyap, R., Kirchner, A., McKay, S., ... McLanahan, S. (2020). Measuring the predictability of life outcomes with a scientific mass collaboration. *Proceedings of the National Academy of Sciences*, 117(15), 8398–8403. <https://doi.org/10.1073/pnas.1915006117>
- Sarma, A., Kale, A., Moon, M. J., Taback, N., Chevalier, F., Hullman, J., & Kay, M. (2021, April). Multiverse: Multiplexing Alternative Data Analyses in R Notebooks. <https://doi.org/10.31219/osf.io/yfbwm>
- Schweinsberg, M., Feldman, M., Staub, N., Van Den Akker, O. R., Van Aert, R. C., Van Assen, M. A., Liu, Y., Althoff, T., Heer, J., Kale, A., Mohamed, Z., Amireh, H., Venkatesh Prasad, V., Bernstein, A., Robinson, E., Snellman, K., Amy Sommer, S., Otner, S. M., Robinson, D., ... Luis Uhlmann, E. (2021). Same data, different conclusions: Radical dispersion in empirical results when independent analysts operationalize and test the same hypothesis. *Organizational Behavior and Human Decision Processes*, 165, 228–249. <https://doi.org/10.1016/j.obhdp.2021.02.003>
- Silberzahn, R., Uhlmann, E. L., Martin, D. P., Anselmi, P., Aust, F., Awtrey, E., Bahník, Š., Bai, F., Bannard, C., Bonnier, E., Carlsson, R., Cheung, F., Christensen, G., Clay, R., Craig, M. A., Dalla Rosa, A., Dam, L., Evans, M. H., Flores Cervantes, I., ... Nosek, B. A. (2018). Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results. *Advances in Methods and Practices in Psychological Science*, 1(3), 337–356. <https://doi.org/10.1177/2515245917747646>
- Simonsohn, U., Simmons, J. P., & Nelson, L. D. (2015). Specification Curve: Descriptive and Inferential Statistics on All Reasonable Specifications. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2694998>
- Simson, J., Pfisterer, F., & Kern, C. (2024). One Model Many Scores: Using Multiverse Analysis to Prevent Fairness Hacking and Evaluate the Influence of Model Design Decisions. *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 1305–1320. <https://doi.org/10.1145/3630106.3658974>
- Starns, J. J., Cataldo, A. M., Rotello, C. M., Annis, J., Aschenbrenner, A., Bröder, A., Cox, G., Criss, A., Curl, R. A., Dobbins, I. G., Dunn, J., Enam, T., Evans, N. J., Farrell, S., Fraundorf, S. H., Gronlund, S. D., Heathcote, A., Heck, D. W., Hicks, J. L., ... Wilson, J. (2019). Assessing Theoretical Conclusions With Blinded Inference to Investigate a Potential Inference Crisis. *Advances in Methods and Practices in Psychological Science*, 2(4), 335–349. <https://doi.org/10.1177/2515245919869583>

- Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*, 11(5), 702–712. <https://doi.org/10.1177/1745691616658637>
- Team, R. C. (n.d.). R: A Language and Environment for Statistical Computing. <https://www.R-project.org/>
- Therneau, T. (2024). *_A Package for Survival Analysis in R_*. R package. <https://CRAN.R-project.org/package=survival>
- Van Dongen, N. N. N., Van Doorn, J. B., Gronau, Q. F., Van Ravenzwaaij, D., Hoekstra, R., Haucke, M. N., Lakens, D., Hennig, C., Morey, R. D., Homer, S., Gelman, A., Sprenger, J., & Wagenmakers, E.-J. (2019). Multiple Perspectives on Inference for Two Simple Statistical Scenarios. *The American Statistician*, 73(sup1), 328–339. <https://doi.org/10.1080/00031305.2019.1565553>
- Wagenmakers, E.-J., Sarafoglou, A., & Aczel, B. (2022). One statistical analysis must not rule them all. *Nature*, 605(7910), 423–425. <https://doi.org/10.1038/d41586-022-01332-8>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., ... Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>